

Question number	Answer	Mark
2(a)(i)	D chemoreceptor, medulla oblongata	(1)

Question number	Answer	Additional guidance	Mark
2(a)(ii)	An explanation that includes any four of the following points: • an increase in blood carbon dioxide decreases the pH of the blood (1) • the pH of the blood needs to be kept within narrow limits (1) • therefore the ventilation rate has to increase (1) • nerve impulse sent from {respiratory centre / medulla} to intercostal muscles (1) • when inspiratory centre is stimulated, the expiratory centre is inhibited (1)	Accept cells have to be supplied with sufficient oxygen / carbon dioxide has to be removed Accept converse	(4)

Question number	Answer	Mark
2(b)	An answer that includes the following points: <pre> graph TD A[This apple releases ethene (1)] --> B[these apples will release more ethene (1)] B --> C[more apples will ripen (1)] C --> B </pre>	(3)

